

Day 2: Diagnosing Hardware FailuresTheory

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
RAM failure	Computer may not boot, may restart repeatedly, or may produce beep codes.
Hard drive/SSD failure	Windows may not start, files may be inaccessible, or the PC may show a “No boot device” error.
Power supply (PSU) failure	PC does not turn on at all, or it suddenly shuts down/restarts.
Graphics card failure	No display, screen artifacts, flickering, or crashes while running graphics-intensive programs.
Overheating CPU	PC becomes very slow, fans run loudly, or the computer shuts down unexpectedly.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Troubleshooting Plan: Laptop Not Charging

- 1. Check the power outlet**
 - Plug another device into the outlet to make sure it is working.
 - **Check:** Whether the outlet is providing power.
- 2. Inspect the charger**

- Check the charger, cable, and plug for cuts, bent parts, or burn marks.
- **Check:** Whether the charger has visible physical damage.
- 3. **Check the charging port**
 - Look inside the laptop's charging port for dust, dirt, or bent pins.
 - **Check:** Whether the port is damaged or blocked.
- 4. **Try another compatible charger**
 - If available, use a known-working charger with the correct voltage and connector.
 - **Check:** Whether the problem is caused by the original charger.
- 5. **Test the charger with a multimeter**
 - Set the multimeter to **DC voltage** and carefully measure the charger's output using the correct terminals.
 - **Check:** Whether the charger is producing the voltage specified on its label.
- 6. **Check the battery connection**
 - If the laptop design allows it, turn it off, disconnect the battery, and inspect the battery connector.
 - **Check:** Whether the battery is loose, swollen, or visibly damaged.
- 7. **Inspect for overheating or burning**
 - Look and smell for signs of overheating around the charger, charging port, and battery.
 - **Check:** Whether there are signs of a damaged component.
- 8. **Test charging again**
 - Reconnect everything securely and plug the charger back in.
 - **Check:** Whether the charging indicator turns on and whether the laptop begins charging.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB Charger Voltage Test

1. Set the multimeter to **DC voltage (V=)**.
2. Connect the **black probe** to the USB charger's **ground (GND)** pin.
3. Connect the **red probe** to the **+5V (VBUS)** pin.
4. Plug the charger into a power outlet.
5. Read the voltage on the multimeter.

Measured voltage: 5.02 V DC

Expected range: 4.75 V to 5.25 V

Result: 5.02 V is within the expected range, so the USB charger is providing the correct voltage.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

12V Rail Test

1. Turn off the PC and disconnect the ATX power connector.
2. Set the multimeter to **DC voltage (V --)**.
3. Put the **black probe on a black wire (ground)**.
4. Put the **red probe on the yellow wire (+12V)**.
5. Read the voltage on the multimeter.

Measured voltage: 12.08 V

Result: The voltage is within the normal range, so the 12V rail is working correctly.

If the voltage is below 11.5V: It could indicate a **faulty or failing power supply (PSU)**, which may cause the PC to shut down randomly.